Lecture XIV - Recap and Discussion

Applied Optimization with Julia

Dr. Tobias Vlćek



Introduction

Congratulations!

You've finished the course and learned how mathematical models can be used to solve real-world problems!

Topics Covered

Question: Can you recall the problem structure behind each topic?

- Solar Panel Transport
- Beer Production
- Split Order Minimization
- Library Routing
- Police Service Districting
- Safety Planning for the Hajj Pilgrimage
- Arena Seating
- Passenger Flow Control

Underlying Problem Structures

Topic	Original Problem[1]
Solar Panel Transport	Classic Transport Problem
Beer Production	Capacitated Lot-Sizing Problem (CLSP)
Split Order Minimization	Quadratic Multiple Knapsack Problem (QMKP)
Library Routing	Capacitated Vehicle Routing Problem (CVRP)
Police Service Districting	p-Median Problem
Safety Planning for the Hajj Pilgrimage	Scheduling Problem
Arena Seating	2D-Knapsack Problem
Passenger Flow Control	Dynamic Network Flow Problem

[1] Original problem structure we **used/extended/exploited** to address the topic.

Modeling Techniques

Question: And do you remember the key modeling ideas?

Topic	Key Modeling Idea
Solar Panel Transport	Sets, parameters and continuous variables
Beer Production	Binary setup variables and Big-M constraints
Split Order Minimization	Quadratic objective with binary variables
Library Routing	Subtour elimination and heuristics

Modeling Techniques II

Topic	Key Modeling Idea
Police Service Districting	Contiguity and compactness constraints
Safety Planning for the Hajj Pilgrimage	Time-indexed scheduling with penalties
Arena Seating	One binary variable per seating group
Passenger Flow Control	Queue dynamics over time periods

What have we learned?

- How to **identify and abstract** real-world problems
- How to **start programming** in Julia
- How to **model and solve** optimization problems
- How to **question** model assumptions

> **Tip**
> That's a lot and a great foundation for a seminar or a master thesis!

Check Yourself

Before the exam, you should be able to answer the following:

- Can you **define sets, parameters, and variables** for a new problem?
- Can you **choose the right variable domain** (continuous, integer, binary)?
- Can you **formulate objective functions and constraints** in JuMP?
- Do you know **when a model needs Big-M constraints**?
- Can you **explain what subtours are** and how to prevent them?
- Do you know the **difference between LP, MIP, and NLP**?

> **Tip**
> If some points feel shaky, revisit the linked lectures and take a look at the cheatsheets on the course website.

Any questions
regarding the
past lectures?

How to continue?

How to continue after the lecture?

- The best way is to **keep programming and modeling**
- We offer **seminar places and master thesis supervision**
- Try to find a way to **apply programming in your work**
- There are **many interesting topics** to explore!

> **Tip**
> Getting your managers on board is the hardest part! But note that it is often worth it. The tools we have used are all free and open-source.

Concrete Next Steps

- Ask questions on the Julia Discourse forum
- Explore the **JuMP community** at jump.dev
- Contribute to open-source, e.g. **"good first issues"** of JuMP
- Start a **small personal project** using the tools learned

> **Tip**
> Project ideas at the right scale: a schedule for your sports league, a weekly meal plan on a budget, or a shift roster for a student job.

Start Pair Programming with AI

- First, try to **be confident with the basics** of a language
- Always try to **understand the code you use**
- Then, try an **AI-assisted IDE or coding agent**

> **Tip**
> The tools change fast; popular options are, for example, Cursor, GitHub Copilot, and Claude Code. They make work much easier compared to copying and pasting code between a chat and your editor.

Final Words

That's it for the Lecture Series!

- I hope you **enjoyed** the lecture and found it **helpful**
- In the last tutorial, we will have a final **discussion session**
- There, you can earn the **last half-bonus point** for the exam
- I wish you **all the best for your studies and your career!**

> **Note**
> If you have any questions on optimization in the future, feel free to contact me!

Questions?

Thank You!

Thank you for participating in the course, and good luck with the exam!

Bibliography